Anomaly Detection of Audio Based on Real Audio Using Generative Adversarial Network Model

ABSTRACT

Audio causes damage not only to individuals and companies, but also to nations; Therefore, research on deepfake audio detection technology is crucial. Most existing deepfake audio detection Research has been conducted using supervised learning; however, when a new synthetic deepfake audio Emerges, real-time detection becomes difficult because of the limitations of supervised learning. Therefore, This paper proposes a new anomaly detection technique for identifying deep-fake audio using unsupervised Learning. This method involves learning the feature distribution of numerous real human voices and then calculating an anomaly score for each voice to determine whether it is deepfake. In this study, we imaged speech Using mel-spectrogram and mel-frequency cepstral coefficient (MFCC), which are speech preprocessing Methods. Subsequently, the parameters of the GANomaly and f-AnoGAN models, which are effective in Detecting abnormalities in speech, were tuned and subjected to unsupervised training. The most effective Result had an F1-score of 0.93 in and was obtained by combining imaging speech with Mel-Spectrogram With training using the GANomaly model.

EXISTING SYSTEM

Current methods of detecting audio forgeries rely on manual analysis or simple feature extraction techniques such as MFCCs or audio fingerprints. These approaches often fail when adversarial attacks are introduced or when deepfake audio is generated with high-quality neural TTS models like Tacotron2 or WaveNet.

PROPOSED SYSTEM

The proposed model utilizes a hybrid deep learning architecture that processes audio by converting it into spectrogram images, followed by classification using Convolutional Neural Networks (CNN). It can distinguish between real and fake audio samples with high precision by leveraging spatial and frequency features embedded in the spectrograms. By transforming audio into spectrograms, the problem becomes an image classification task, enabling the model to detect subtle patterns often missed in raw waveforms.

PURPOSE OF THE PROJECT

The current system for detecting deepfake audio using spectrogram analysis and deep learning has proven effective, but it offers significant potential for further development. One of the most impactful future enhancements would be supporting multilingual audio detection, allowing the system to identify fake content across different languages and dialects.

ADVANTAGES

High detection accuracy
End-to-end automation
Spectrogram-based feature analysis
Easy-to-use GUI
Scalable for real-world use
Robust across various audio types

MODULES

The application comprises the following modules

Audio Upload Module
Spectrogram Generator
Feature Extractor
Classifier Model
Prediction Module
Graphical UI

SYSTEM CONFIGURATION/REQUIREMENTS

HARDWARE REQUIREMENTS

Processor	intel i3(min)
Speed	1.1 Ghz
RAM	256 MB(min)
Hard Disk	20 GB

SOFTWARE REQUIREMENTS

Operating System	Windows 10(min)
Programming Language	Python
Development Environment	Jupyter Notebook / VS Code
Libraries	Tensorflow, Numpy, Tkinter,